EFU NITROGEN FRAMEWORK v2.0 – RESEARCH AUDIT EDITION

Identifier: EFU-A/NITROGEN-2026-01-R01
Status: Research Material – Open for Critical Review (Research Phase)
Author: István Simor (ORCID: 0009-0002-6599-3480)
Audited by: EFU Scientific Coordination Unit
Date: January 2026 (derived from Original Standard draft: 2025-12-30)
Licence: EFU Universal Licence v1.0 (CC BY 4.0 compatible)

ABSTRACT

This research document reformulates the EFU Nitrogen Standard as a scientific model under audit conditions. It treats the **3.2 kg N person^{-1} year^{-1}** baseline not as a regulatory value but as a **flux reference hypothesis**.

The aim is to verify whether expressing industrial, agricultural, and ecological nitrogen fluxes in **human-equivalent EFU-N units** increases transparency, comparability, and communication of nitrogen footprint data.

This edition aligns with the EFU ecological flux methodology and emphasises **falsifiability, uncertainty disclosure, and cross-element interpretation** (carbon–nitrogen coupling).

---

0. Research Status and Model Scope

[HYPOTHESIS]
Human metabolism provides an intuitively stable scale for interpreting the magnitude of global nitrogen fluxes.

[PROTOCOL]
The *EFU-N* unit is proposed exclusively for **research and comparative metrics**, not as a legal or certification standard.

The model applies to agriculture, wastewater, industrial emission studies, and integrated environmental accounting.

I. Definition of the Unit and Scientific Basis

[AXIOM]
Nitrogen obeys mass conservation; thus, total anthropogenic reactive nitrogen accumulation reflects **input–output imbalances** of a closed planetary nitrogen cycle.

[HYPOTHESIS]
1 EFU-N = 3.2 kg N year^{-1} person^{-1} captures the mean human nitrogen turnover corresponding to protein metabolism.

[PROTOCOL]
For cross-scale comparison, all nitrogen mass or compound flows (NH_3, NO_3^-, N_2O, urea) are converted to **kg N** before normalisation.

EFU-N Conversion Formula

EFU-N = mass of nitrogen (kg N yr^{-1}) / 3.2

Reference Conversion Fractions

Compound	N fraction	Conversion example
Ammonia (NH_3)	0.824	EFU-N = kg NH_3 × 0.824 / 3.2
Nitrate (NO_3^-)	0.226	EFU-N = kg NO_3^- × 0.226 / 3.2
Nitrous Oxide (N_2O)	0.636	EFU-N = kg N_2O × 0.636 / 3.2
Urea [$CO(NH_2)_2$]	0.467	EFU-N = kg urea × 0.467 / 3.2

II. Anthropocene Context (Hypothetical Analysis)

[HYPOTHESIS]
Technospheric production of reactive nitrogen (Nr) **(150–210 Tg N yr^{-1})** exceeds the biospherically tolerable boundary **(~60–80 Tg N yr^{-1})** by a factor of **2–3**.

[ARGUMENT MAP]

- **A:** Global metabolic nitrogen flux ≈ 25–30 Tg N yr^{-1} (8 billion people × 3.2 kg).
- **B:** Artificial Nr production ≈ 200 Tg N yr^{-1}.
- **C:** Technospheric flux ≈ 6× metabolic flux → planetary imbalance detectable.

This scale interpretation enables a **human-centred expression** of systemic overproduction.

III. Sectoral Prototyping (Research Models)

Agriculture

[PROTOCOL]
Nitrogen Use Efficiency (NUE) may be represented in EFU-N terms:

NUE = EFU-N (harvested product) / EFU-N (fertilizer input)

Target (**NOT a standard**): ≈ 0.6 for balanced agro-ecosystems.

Wastewater Systems

[HYPOTHESIS]
Advanced treatment can reduce per-capita nitrogen discharge to **< 0.2 EFU-N person^{-1} yr^{-1}** by 2030.

[PROTOCOL]
Results should report uncertainty (±10 %) and boundary conditions (inflow composition, temperature, flow rate).

Industrial / Ammonia Sector

[HYPOTHESIS]
Global NH_3 output ≈ 180–200 Mt yr^{-1} ≈ 150–165 Mt N ≈ O(40 million EFU-N).

[PROTOCOL]
Use standardised molecular fractions for conversion. All values are **communicative, not normative**.

IV. Audit and Reporting Model

[PROTOCOL]
Entities adopting EFU-N for research should explicitly state:

- applied conversion formulas and speciation assumptions;
- distinction from conventional units (kg N, CO_2-eq, etc.);
- spatial and temporal system boundaries;
- uncertainty margin (%) based on data tier (1–3).

Illustrative disclosure example:

"Annual Nitrogen Flux = 15 625 EFU-N (±10 %), calculated using EFU-N Research Framework v1.0 (Simor, 2026)."

V. Epistemological Appendix (EA)

Falsifiability

[HYPOTHESIS]
The hypothesis fails if the 3.2 kg N baseline shows no robust correlation between dietary intake and nitrogen excretion across demographics ($R^2 < 0.3$).

It also fails if planetary Nr flux data (150–210 Tg N yr^{-1}) are found to be comparable to biospheric sink capacity without overshoot.

Uncertainty Factors

- regional variability in protein intake and N metabolism;
- nitrogen speciation errors and molecular fraction approximation;
- incomplete data on wastewater removal efficiency;
- conversion of organic to reactive forms.

Terminology (Context-Specific)

- **Fluxus:** dynamic mass transfer per unit time.
- **Metabolic Dignity:** right of systems to sustain minimal biogeochemical flux for functionality.
- **Sovereignty Gap:** disparity between technospheric production and biospheric assimilation capacity.
- **Audit:** scientific assessment for falsification and model validation, not certification.

VI. Critical Argument Map Audit

#	Research Statement (Hypothesis)	Empirical Basis / Logic	Possible Vulnerability	Suggested Falsifier
1	Human metabolic N turnover ≈ 3.0–3.5 kg yr^{-1}	WHO/FAO nutrition data	Dietary variance (~±30 %)	Urinary N field study by region
2	1 EFU-N can scale global Nr flux to human reference	Dimensional analysis	Anthropocentric bias	Cross-validation with planetary mass-balance
3	Technospheric N > biospheric N need × 6–8	Global flux inventories	Model uncertainty > ±25 %	Recalculated global N-cycle data
4	Planetary safe boundary ≈ 60–80 Tg N yr^{-1}	Steffen et al. 2015	Threshold definition debated	Model update with new boundary papers
5	EFU-N fosters "human-equivalent" communication	Comparative notation logic	Oversimplification risk	Stakeholder comprehension survey
6	NUE ≥ 0.6 represents efficiency target	FAO data	Crop-specific divergence	Multicrop NUE meta analysis
7	WW treatment can achieve < 0.2 EFU-N/person	Current tech benchmarks	Operational variability	Pilot plant data
8	NH_3 production ≈ 40 million EFU-N	FAO/IEA statistics	Data range wide	Independent industry audit
9	EFU-N allows cross-element integration (C ↔ N ↔ P ↔ W)	EFU framework design	Data heterogeneity	Multi-element correlation test
10	Human flux ratio to Nr reveals overshoot	Flux ratio analysis	Population growth sensitivity	Projected demographic scenarios
11	EFU-N metric enhances transparency	Audit logic	Interpretation complexity	User data compliance survey
1	Compound-to-N fraction stable	Atomic weights	Temperature and state e	Laboratory mass-fraction validation

#	Research Statement (Hypothesis)	Empirical Basis / Logic	Possible Vulnerability	Suggested Falsifier
2			ffects	
13	Falsification possible via isotopic tracing	15N studies	Cost and resolution limits	Isotopic field trial
14	Integrating EFU-N reduces “greenwashing”	Comparative audit	Data-openness dependence	Transparency benchmark survey
15	Planetary imbalance quantifiable in EFU terms	Flux normalisation	Baselining assumptions	Sensitivity analysis
16	EFU-N can guide policy communication	Educational scaling	Political interpretation bias	Policy-impact pilot
17	Open licensing supports scientific replication	CC BY 4.0 framework	Attribution tracking	Citation audit
18	Sovereignty Gap metric applicable to N	EFU terminology	Boundary uncertainty	Regional flux gap estimation
19	Metabolic Dignity applies to biogeochemical systems	Analogy from EFU-HSW	Metaphoric ambiguity	Operational definition test
20	EFU-N scalable to enterprise audits	Modular architecture	Commercial-use ambiguity	Pilot corporate audit

---

VII. Summary Interpretation

This research-phase text treats EFU-N as an **open analytical language** for planetary nitrogen dynamics. All quantitative values are provisional and subject to empirical adjustment.

The framework’s scientific merit depends not on normative adoption but on **measurable falsification and transparent uncertainty disclosure**.

This document represents an ongoing research construct; all models and thresholds are provisional and subject to revision or falsification.

AI-assisted tools were used for linguistic refinement and structural consistency under full human conceptual control.